

Neurally-Inspired Gesture Recognition

- What are gestures? → A movement of a limb or the body as an expression of thought or feeling
- Why are gestures relevant?
 - ↳ But: In different cultures different meanings
 - visual channel in multimodal communication
 - interdisciplinary research (AI, psychology, neuroscience)
 - Human ~~Robot~~ Computer Interaction (HCI): input, navigating
 - Human Robot Interaction (HRI):
 - passive: responsive agent
 - active: learning of gestures / concepts

Gesture Continuum

- different abstraction levels in gesture meaning
- increasing level of linguistic properties and semantics

"Peace"

Gesticulation → Iconic → Deictic → Pantomimes → Emblems → Sign language

non-verbal: accompanying body expression

Communicative: supports or replaces speech

- Data acquisition for gesture recognition:

↳ sensors: state of hand; etc.

Towards vision-based gesture recognition

- most intuitive →
 - (+) no cables
 - (+) communication at distance
 - (+) sign language

→ needs

- Skeletal Data
- depth sensors!
- huge processing

Issues:

- (-) Env. influence: Background, Distance to sensor
- (-) Sensor choice: 2D, 3D, Noise
- (-) Single user / Multiple user
- (-) Task

• Static and Dynamic Gestures

→ Static Gestures:

→ static postures, no temporal information to be recognised

→ eg: "OK" 1/5

→ Important: hand shape and finger configuration

→ Challenge: postures with complex backgrounds

→ Models: → Template-Matching, Elastic Graph Matching
→ SVM, MLP, 2D/3D CNN

→ Dynamic Gestures

→ dynamic hand and arm movement and trajectories

→ eg: "move to the left"

→ Important: Spatiotemporal pattern recognition

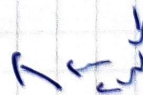
→ Challenge: start and end of isolated or continuous gestures

→ Models: → Probabilistic graphical models (HMMs)
→ RNN, SOM growing when required networks

• Gesture Representation

→ Appearance-based:

→ Hand Contours
→ Image Moments
→ Optical Flow



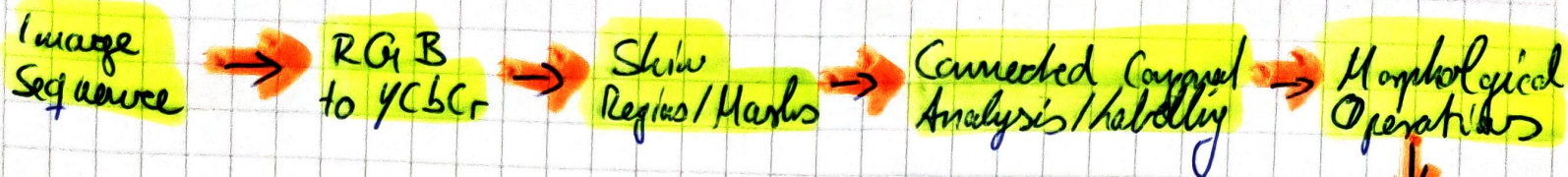
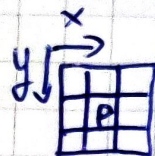
→ Model-based:

→ Joints to Model finger movements
→ Mesh grid
→ Geometric hand model



L10

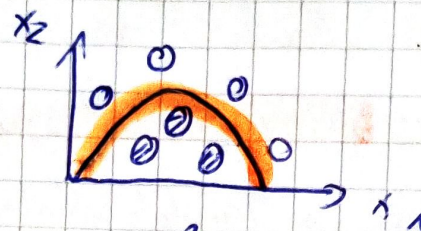
Image Processing Pipeline for gestures



Threshold: Hand

$$y = f(\sum w_i x_i)$$

Neural Networks



$\rightarrow$ Decision Boundary:

$\rightarrow$ How to compute decision boundary?

$\rightarrow$ Minimization of network error

$\rightarrow$ Objective function usually mean-squared criterion

$$E_{\text{net}} = \frac{1}{2} (y - \bar{y})^2$$

$\rightarrow$ Learning = weight update

$\rightarrow$ f nonlinear for arbitrary shapes

$\rightarrow$ continuous and differentiable (e.g. sigmoid)

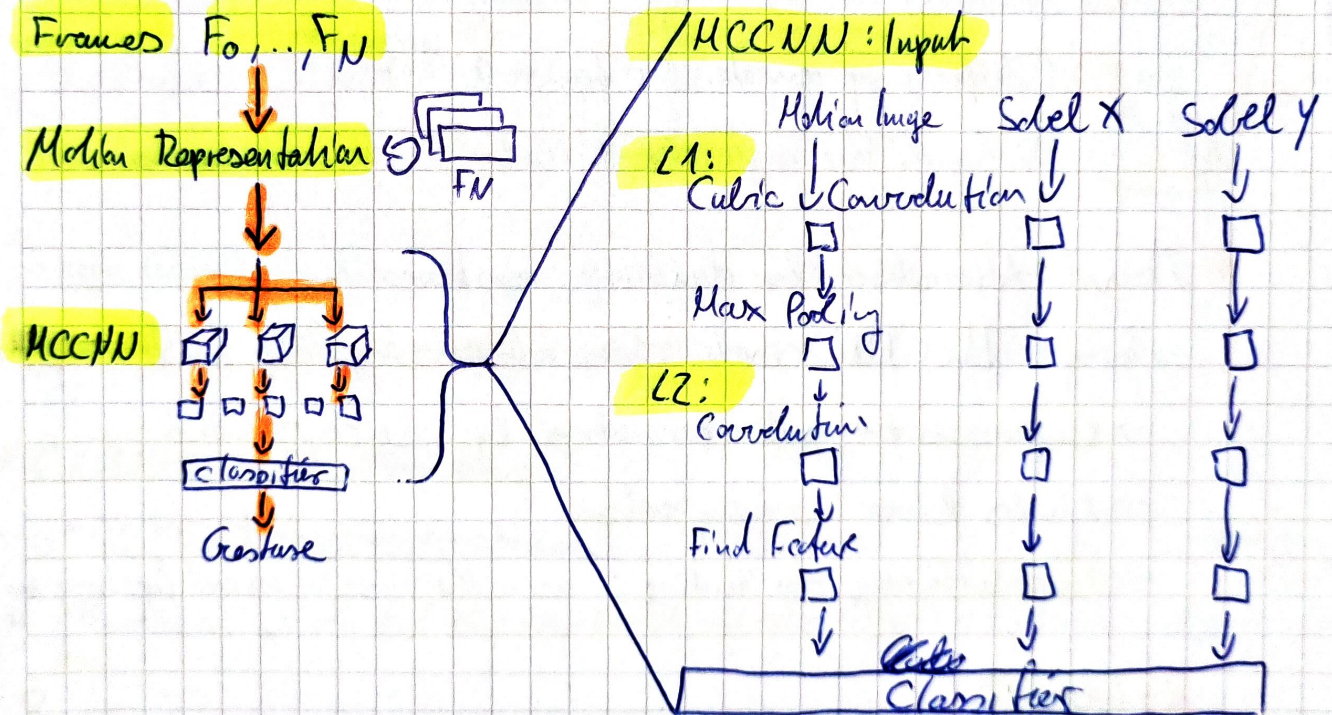
$$f(x) = \frac{1}{1 + \exp(-x)} \quad , \quad \frac{\partial}{\partial x} f(x) = f(x)(1 - f(x))$$

$\rightarrow$ Error backpropagation w.r.t weights $\frac{\partial E_{\text{net}}}{\partial w_i} \Rightarrow$ Gradient

L10

• Multichannel Convolutional Neural Network (MCCNN)

- Extension to CNN with 3D Kernel
- Convolution and pooling along image stacks in 1st layer
- Application of Sobel Filters
 - Edge detection along the horizontal and vertical image direction
- Can be used for temporal data
 - Learning the shape of the Motion History Image (MHI)
 - extended application to dynamic gestures



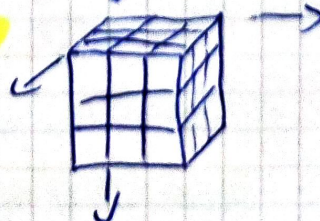
1D Convolution Kernel



2D Convolution Kernel



3D Convolution Kernel



• Evaluation for alternatives for Deep NN (DNN)

→ DNN:

- (+) Successful benchmarking in vision and audio domain (+)
- (+) Robust feature emergence ^{Entstehung} due to hierarchical processing
- (+) Optimization efforts in open software
- Trends regarding transfer learning and generative or unsupervised models
- (-) Data - Hungry and Time - Consuming
 - Exhaustive tuning and retraining, and optimization strategies
- (-) Specialist for one task (Adaptability & Feedback?)
- (-) Not fully appropriate for temporal domain
 - 3D kernel no significant advantage over 2D kernels

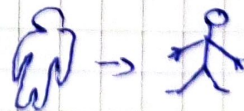
• Neural Alternatives for dynamic gestures

- Simple MCCNN arch does not yet explicitly model temporal Domain
- Compresses temporal information by 3D convolution
- Needs large training sets
- Computationally demanding: ^{→ do not always meet real-time conditions} as is the case in general for training with deep NN

⇒ Idea:

~~SCD~~

→ Benefit from new preprocessing technology:



- Skeleton data provides necessary body joints
- Use dynamic recurrent gesture representation
 - Model with unsupervised self organizing maps or LSTM
- Meet real-time condition

LSTM = Long - Short - Term Memory

Hybrid CNN and LSTM for Dynamic Gestures

→ Datasets: Commands for HRI and ChaLearn benchmarks

→ CNN:

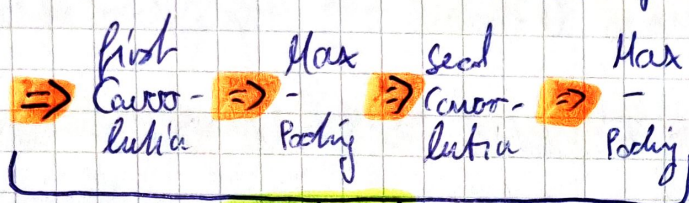
→ Learning of significant visual features

→ Invariance spatial properties

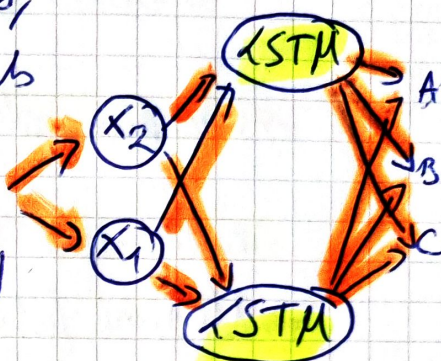
→ LSTM:

→ Variations in performance: Gesture Types, Across Subjects

Differential Image Sequence



CNN



→ Good performance for gestures with motion profile

→ Decrease for more subtle gestures and ChaLearn Gestures

CNN LSTM Variations

→ can be tuned in various ways due to its cell state

(one-to-one config)

→ Prediction at the last timestep (many-to-one config) vs prediction at each timestep

Model	Frame-level CNN LSTM	Sequence-level CNN LSTM
Training Method	online training	mini-batch training
Data preparation	supports varying-sized sequences	requires slicing and padding to a fixed length
Classification	requires post-processing to produce sequence classifications based on frame predictions	one classification per gesture sequence

Hybrid Networks for Dynamic Gesture Recognition

- How to deal with similar movement patterns and subtle movement
- Extension of the CNN+STM arch:
 - Hybrid (static/dynamic) gesture recognition
 - Combine temporal modeling with the posture
 - modular & lightweight arch
 - evaluation on multiple gesture domains: word commands and co-speed
- Gesture phases (Kendon) , Spirophoto peaks

Rest Position → Pre-Stroke → Stroke → Post-Stroke → Rest Position

→ Motion Intensity Profile

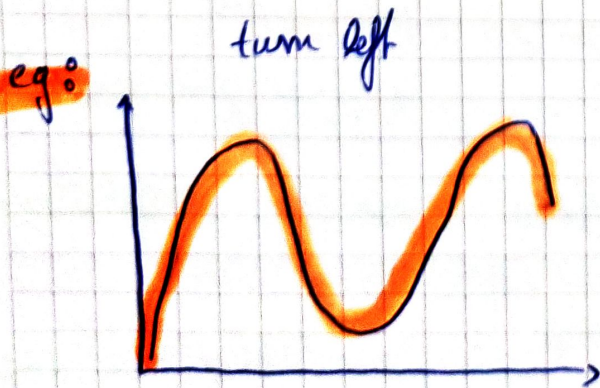
- Challenge: how to analyze motion/pause carried in a movement sequence?
- Solution:

→ Structure Similarity (SSIM) index calculates the similarity between two frames using luminance, contrast and structure

→ SSIM can be inverted to find the change in motion across a timespan: $ISSIM = \text{inverted SSIM}$

$$ISSIM = 1 - SSIM(t_i, t_0)$$

→ The first frame of the sequence is used as a reference



Hybrid Network: Arch of the Snapshot Model

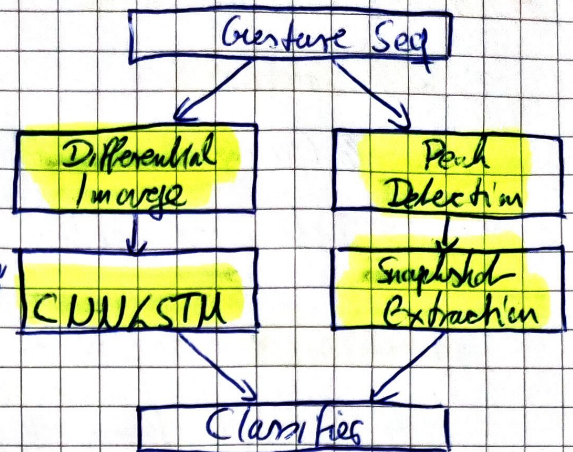
→ Static & Dynamic Channels

→ Input: Isolated Gesture Sequence

→ Classification boost of indistinctive movements

→ Classification boost of subtle movements → "feinstrukturel" → enhanced HRI

→ Limitations: Gestures with similar motion and hand pose



Growth - When - Required Networks (GWR networks)

→ Extension based on self-organizing maps (SOM)

→ SOM-size input-driven and not fixed

→ Grow / Shrink by adding/removing nodes

→ Grow and add new nodes depending on distance to input vector

→ Shrink by removing nodes that do not have any connections to neighbours anymore

GWR-Alg
→

• GWR - Algorithm

→ Initialization :

- Create two nodes randomly initialized from the input
- Define connection^{set} between the nodes : empty

→ Iteration :

- Sample from Input Data
- Calculate (Euclidean) distance for each node to sample :

$$d_j = \|x(t) - w_j\|$$
- Select best-matching unit (BMU) and second BMU (SBMU)
- Create connection with age 0 between BMU & SBMU if none-existent
- Calculate activity of BMU

$$a(t) = \exp(-d_{\text{BMU}})$$
- IF activity and habituation ^(firing rate) < threshold : $a(t) < a_T$ & $h_b < h_T$
 - Add new node
 - Set weight of new node
 - Create connection between BMU and new node & SBMU and new node, removing the connection between BMU & SBMU
- Else update neighboring nodes

$$\Delta w_i = \epsilon_i \cdot h_i \cdot (x(t) - w_i)$$

$$\Delta h_i = \tau_i + 1.05 \cdot (1 - h_i) - \tau_i$$
- Increment age of all connections ending in the BMU
- Reduce firing counter for BMU and neighbors
- Check for removable connections and/or nodes

Gamma GWR

→ Extension of GWR-dlg

→ Add **recurrences** to GWRs

→ Nodes do not only contain a weight vector, but also a **context vector**

→ **Context descriptor** is introduced

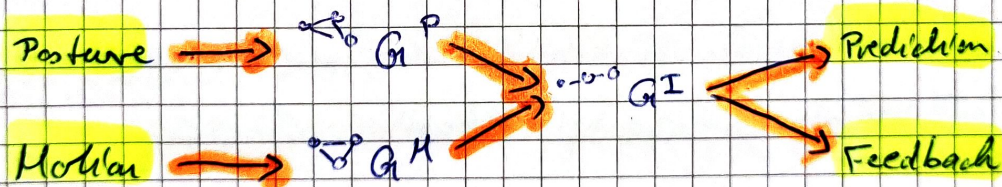
→ Combination of previous BMU's weight and context vector

→ **Distance** between input sample and node is now determined not only by the weight but also the **context**

→ BMU's of previous time steps influence the current selection of the BMU

→ This allows for the learning of **temporal sequences**

Motion prediction for physical exercises



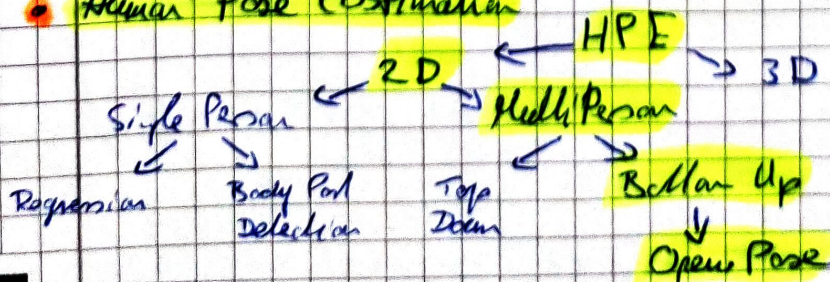
→ Use **Gamma GWR** to learn **pose** in a physical exercise

→ Predict **next pose** in a sequence that defines the physical exercise

→ **Compare pose with prediction** and calculate error between joints

→ Joint positions by **Human Pose Estimation (HPE)** framework **OpenPose**

Human Pose Estimation



Open Pose - Arch of multi-stage CNN



L : Part Affinity Fields Maps

F : Feature Maps

p : Loss after each stage

S : Part Confidence Maps

ϕ : CNN of Branch 1

p : CNN of Branch 2

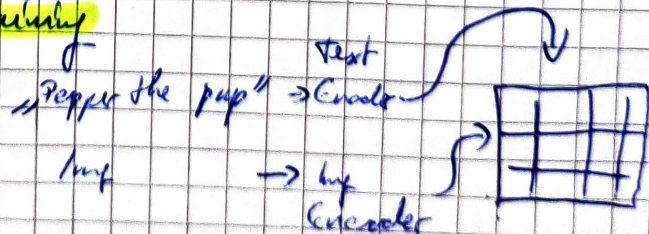


GLOR for polarity in HRI

- Growing-When-Required Network shows robust classification performance, even when gesture is ambiguous
- Solves ambiguity when polarity
- polarity gesture too far from object, therefore not recognized as relevant
- if ambiguous, closest possible object location returned

Contrastive Language-Image Pretraining

→ (1) Contrastive pre-training



→ (2) Create Dataset classifier for label text

"A photo of 53"

→ (3) Use for zero-shot prediction

img →  → A photo of a dog



Summary

- Gestures are an integral part in communication
- Important in multimodal systems
- Research on autonomous robots and HRI
- Model alternatives and choice
 - Dependent on gesture type → gesture continuum
 - Deep NN provide partial solution for feature extraction
 - RNN in LSTM and GANNA GWR
 - Reservoir Computing models by neurophysiological principles for action selection in prefrontal cortex and striatum
- Tradeoff between benchmarking and developing cognitive systems